Reads Right, Judges Wrong: Near-Miss Failures and the Limits of Uncertainty Recovery in VLM Success Detectors

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***Abstract*—Robot learning pipelines use vision-language models (VLMs) as success detectors and reward models: the model is told what the robot was instructed to do and asked whether the scene shows it done. The informative cases are near-misses, where the outcome almost matches the instruction. We study them on a controlled benchmark of robot tabletop arrangements with exact ground truth (colored blocks or letter tiles, instructed into a memorized order such as the alphabet or into a random one) and on 200 real TACO-Play episodes, each judged against its true instruction and against a counterfactual that changes one detail. The detector reads right and judges wrong: Qwen3.5-4B passes 71 of 144 arrangements with a single adjacent swap while failing 0 of 96 correct ones, yet in a separate call it transcribes 70 of the 71 swaps it passes correctly. The discrepancy also appears within one response: Qwen3.5-0.8B lists a failed arrangement exactly and then answers PASS in 122 of 122 such answers. For the strongest model the error is recoverable. Comparing a neutral transcription in code removes it, and its verdict score ranks failures above successes (AUROC 0.98), so a conformally calibrated threshold holds false PASS at 9% over all failures with 0% false FAIL. That pooled guarantee does not cover each failure type: under it 21% of swaps still pass, and 9% (at 8% false FAIL) when the threshold is calibrated on swaps. Weaker models add a harder failure, as instructions and memorized orders leak into what they read. On real robot frames, under a final-frame protocol, the recovery does not transfer: verdict scores show weak pooled discrimination (AUROC 0.45–0.51), and Qwen3.5-0.8B and 4B certify both of two mutually exclusive outcomes for the same frame (such as opening and closing a drawer) on 100% and 74% of frames.**

I. Introduction

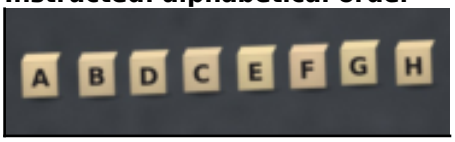
A robot that checks its own work needs a model that can say "this is not what I was asked to do." Vision-language models (VLMs) are now routinely placed in that role: as success detectors for manipulation [1], as zero-shot reward models [2, 3], as failure explainers [4], and as the perception front end of vision-language-action policies [5]. In each case the model is conditioned on an *expectation*, the instruction or goal, and asked to compare it with an *observation*. For a success detector the informative cases are the near-misses, where the outcome almost matches the instruction: one object in the wrong place, the right action on the wrong item. A detector that is right only when the robot clearly succeeded or clearly failed adds little.

Viewed as belief updating, a success detector should combine a prior (what it was told to expect, and what it already knows about such scenes) with a likelihood (the pixels). We ask how the balance is struck on near-misses, and whether the model represents the conflict in its uncertainty. The question matters for uncertainty-aware robotics: if a VLM's verdict score ranks its errors, tools such as conformal prediction [6, 7] can turn it into decisions with a controlled error rate under stated assumptions; if failures receive scores as low as successes, those tools inherit the blind spot. We study the success-evaluation component of robot learning systems, the part that labels outcomes for evaluation, data filtering and reward, rather than predictive dynamics or how uncertainty propagates through planning.

We build two benchmarks around near-misses. A controlled one renders robot tabletop arrangements with exact ground truth in a 2×2 design: colored blocks or letter tiles, instructed to follow either a memorized order (rainbow, alphabetical) or a random one. A real-robot one pairs the final frames of 200 TACO-Play episodes with their true instruction and with a counterfactual that changes a single detail. We evaluate Qwen3.5 at three sizes and Gemma 4 E2B. Our findings:

- **The detector reads right and judges wrong.** Asked whether the robot succeeded, Qwen3.5-4B passes 49% ([41, 57]%) of arrangements with a single adjacent swap while failing 0 of 96 correct ones. This cannot be explained solely by an inability to read these scenes: in a separate call it transcribes 70 of the 71 swaps it passes correctly (Fig. 1). Every model we test passes many near-misses.
- **The discrepancy appears within a single response.** Asked to list the items and then judge, Qwen3.5-0.8B writes down the failed arrangement exactly and still answers PASS in 122 of 122 such answers (Qwen3.5-4B: 7 of 323), and the failure persists when the verdict wording is reversed.
- **For the strongest model, explicit comparison or a recalibrated threshold recovers it.** Comparing a neutral transcription in code is error-free for Qwen3.5-4B. Its verdict score ranks failures above successes (AUROC 0.98), and a conformally calibrated threshold holds false PASS at 9% over all failures with 0% false FAIL; the pooled guarantee leaves 21% false PASS on swaps, and calibrating on swaps brings it to 9% at 8% false FAIL. Weaker models are harder to recover, because stated instructions and memorized orders leak into what they read: three of

Instructed: alphabetical order



instruction	A, B, C, D, E, F, G, H
in the image	A, B, [D], [C], E, F, G, H
neutral prompt	A,B,D,C,E,F,G,H
order given	A,B,D,C,E,F,G,H
PASS/FAIL	PASS (verdict score 0.33)

Instructed: a random order



instruction	F, A, B, C, E, G, H, D
in the image	F, A, [C], [B], E, G, H, D
neutral prompt	F,A,C,B,E,G,H,D
order given	F,A,C,B,E,G,H,D
PASS/FAIL	PASS (verdict score 0.28)

Alphabetical, blank tiles



instruction	A, B, C, D, E, F, G, H
in the image	(no letters)
neutral prompt	A, B, C, D, E, F, G, H
order given	A, B, C, D, E, F, G, H
PASS/FAIL	PASS (verdict score 0.15)

Fig. 1. Qwen3.5-4B as a success detector (greedy decoding). Positions in [brackets] deviate from the instruction. Green: the answer matches the image; red: it does not. Top two rows: asked to list the tiles, the model reads the swap correctly with or without the instruction in the prompt, yet asked whether the robot succeeded it answers PASS, for an alphabetical and for a random instruction alike. Bottom: on blank tiles it reports the alphabet, the order it has memorized.

the four read swapped alphabetical tiles as sorted even with a neutral prompt.

- **The recovery has limits on real frames.** Under a final-frame protocol on TACO-Play, every model's verdict score shows weak pooled discrimination, so thresholding gives little benefit; two models certify both of two mutually exclusive outcomes for most frames, and descriptions of the same frame often change with the instruction.

II. Related Work

VLMs as success detectors and reward models. VLMs have been used to detect task success [1], to provide dense or sparse rewards [2, 3], and to explain failures [4]. Recent benchmarks probe their reliability on real robot data: RoboReward builds negatives and near-misses by counterfactually relabeling successful episodes [8], FailBench finds VLM judges near chance when the outcome depends on subtle visual evidence and biased toward predicting success when evidence is ambiguous [9], and ROBORMBENCH shows that paraphrasing an instruction can flip a VLM reward between success and failure [10]. Our real-robot protocol follows the same counterfactual idea. What we add is a controlled setting that isolates three things these benchmarks do not: whether the model can read the scene (a transcription measured on the same images), whether the expectation is merely instructed or also memorized, and whether the verdict score still ranks failures well enough for a calibrated threshold to recover them.

Priors and context overriding perception. VLMs miscount features of famous objects when the image is edited [11], fail simple visual tests [12], and hallucinate from language priors [13, 14]. Language models resolve conflicts between context and parametric memory inconsistently [15, 16] and tend to agree with what the prompt implies [17]. Our design separates the two sources: on a given image only the prompt changes, and memorized and arbitrary expectations are compared across scene families built with the same generator.

Uncertainty for robot decision making. Conformal prediction gives distribution-free guarantees [6, 18] and has been used to make LLM planners ask for help [7]. Such guarantees are only as useful as the score they threshold: models often, but not always, know what they know [19, 20]. We test whether VLM verdict scores carry the near-miss signal.

III. Benchmarks

Controlled benchmark: a 2×2 design. A robot arm was told to arrange items on a table in a given left-to-right order, and we render what it might have produced (Fig. 2). There are two visual domains, six colored cubes drawn from eight colors and eight wooden letter tiles, and two kinds of expectation per domain. A *memorized* expectation is an order the model knows by heart: rainbow order (red, orange, yellow, green, blue, purple) for blocks, alphabetical order (A–H) for tiles. An *arbitrary* expectation is a random order of the same kind of items, drawn independently for every image, which the model can know only from the prompt. Within a domain the two families share the renderer, the nuisance settings and the prompt templates, but their images are rendered independently, so comparisons between families are not on identical images. Tile goals in both families use A–H; random block goals draw six of the eight colors, so unlike rainbow goals they can include pink



Fig. 2. Controlled benchmark: one example from each family. A robot arm was told to arrange the items in the listed order; the image shows what is on the table.

or white, which makes color composition a possible confound in the block comparison. For each of the four families and each of six imaging conditions (neutral, warm ∼3000 K and green fluorescent illuminants with a fixed white balance; near and far) we render correct arrangements, every adjacent swap of the instructed order, single substitutions (a wrong block, one of the unused colors; a wrong tile, a letter from I–P), and random permutations with at most one fixed point. Swaps and substitutions are near-misses; random permutations are gross failures. Three blank scenes per family (gray blocks; tiles with no letters) carry no information about the order. Blur, sensor noise, JPEG compression and small rotations are randomized. In total 444 images with exact labels.

Prompts. Every image is queried under seven prompts that differ only in text. Four ask for a transcription: *none* ("6 colored blocks / 8 letter tiles in a row"), *scene only* ("a table after a robot arm has arranged …"), *mentioned* (the robot was instructed to use rainbow order, alphabetical order, or "a specific order", without listing it), and *given* (the instructed order is listed, as a task specification would). Allowed color names are listed alphabetically so the list never leaks rainbow order; tiles get no list, since an alphabetical list of letters would be the expectation itself. Three prompts ask for a verdict with the order given: *PASS/FAIL*; the reversed question *"is any item out of place?"* (YES = failure), a control for bias toward an answer word; and *list first*, which asks for the list and then the verdict in one call. The appendix gives every prompt verbatim.

Real-robot benchmark. To test the pattern on real robot images we use TACO-Play [21, 22], a Franka Panda tabletop dataset of language-annotated segments, through its Open X-Embodiment release [23] in the LeRobot format [24]. From the static camera we take the last frame of 40 randomly sampled segments for each of five instruction types whose outcome is visible in that frame: opening or closing a drawer, sliding a door left or right, switching a light on or off, placing a block on the table or on top of the drawer, and grasping a block of a given color. Each frame is judged twice with identical wording: against its own instruction, which the robot completed (correct verdict PASS), and against a near-miss counterfactual that changes exactly one attribute (open↔close, left↔right, on↔off, table↔drawer top, or the block's color), for which the correct verdict is FAIL (Fig. 5). The 200×150 frames are upsampled 4× (Lanczos). Three prompts mirror the controlled ones: PASS/FAIL, the reversed question, and *describe first*, a one-sentence description of the relevant objects followed by the verdict.

Models. Qwen3.5 at 0.8B, 2B and 4B parameters [25], natively multimodal models from the same architecture line as Qwen's current open frontier models, and Gemma 4 E2B [26] (2.3B effective parameters, 5.1B with embeddings) as a second model family, at its default 280-token image budget. All run in bf16 with greedy decoding and reasoning disabled. A reasoning-enabled subset for Qwen3.5-4B (the swaps under neutral light, near) uses Qwen's recommended sampling settings. The appendix lists the model revisions, generation limits and every prompt, including the real-robot ones.

Metrics. A verdict is scored by its false PASS rate on failed arrangements, its false FAIL rate on correct ones, and Youden's J (detection rate minus false-alarm rate; $J = 0$ indicates no discrimination at this operating point, $J = 1$ is perfect). An answer with no parseable verdict does not certify success and is scored as FAIL; we report coverage, the share of answers with a verdict, where it is below 100%. For transcriptions of adjacent swaps we score whether both swapped positions are read correctly or both are reported in the instructed order. The *verdict score* is a normalized token-prefix score: at the step where the verdict is emitted, the total probability of the tokens that can begin the word FAIL, divided by that of the tokens that can begin either verdict word. It is a ranking score, not the probability of a complete answer. Conformal risk control [18] picks, on a calibration split, the $\lfloor \alpha(n+1) \rfloor$-th smallest score among the n calibration failures as the FAIL threshold; under exchangeability of calibration and test items this bounds the expected miss rate by $\alpha = 0.1$ for the failure distribution of the calibration set, here the pooled mix of failure types, not for each type separately. We average 50/50 splits over 500 repetitions; these resplits share items and are not independent datasets. Intervals are 95% Wilson intervals that treat images as independent.

IV. Results

A. The detector reads right and judges wrong

With a neutral prompt Qwen3.5-4B names every block and tile correctly on random sequences in all four families (100%, 100%, 100%, 100%) and reads 99% of adjacent swaps correctly. Asked whether the robot followed the instruction, it answers PASS on 29% of failed arrangements and FAIL on 0 of 96 correct ones ($J = 0.71$, pooled over the four families). The errors are concentrated on near-misses: it passes 71 of 144 single adjacent swaps (49%, [41, 57]%) and 27% of single

wrong items, but only 1% of random rearrangements. Of the 71 swaps it passes, it transcribes 70 correctly when simply asked to list the items in a separate call ([92, 100]%; Fig. 1). The failure therefore cannot be explained solely by an inability to read these scenes, although a separate call does not show which internal evidence the verdict call used.

The same discrepancy appears within a single response. With the list-first prompt, Qwen3.5-0.8B writes down the failed arrangement exactly and then answers PASS in 122 of 122 such answers, and Qwen3.5-4B in 7 of 323.

The failure persists when the verdict wording is reversed. Asked the reversed question ("is any item out of place?", YES = failure), it still misses 38% of swaps while flagging 0% of correct arrangements ($J = 0.83$). The failure also persists when instruction-conditioned transcription is accurate: with the instruction in the prompt the transcription still reads 94% of swaps correctly. What changes the verdict is making the model look before it judges. Asking it to list the items and then answer in the same call reduces false PASS on swaps to 12% ($J = 0.94$), and comparing a neutral transcription against the instruction in code, with no model verdict at all, is error-free ($J = 1.00$). Table I gives the breakdown by family.

B. The verdict score ranks failures, so thresholding recovers them

The verdict is wrong, but its score is informative (Fig. 3). Qwen3.5-4B's verdict score separates failed from correct arrangements with AUROC 0.98 (pooled; 1.00, 1.00, 0.99, 0.98 by family): its mean score is 0.47 on swaps and 0.72 on wrong items, against 0.04 on correct arrangements, but greedy decoding effectively thresholds it at one half (the emitted verdict agrees with a score above 0.5 on 99.5% of answers). Moving the threshold is enough. Conformal risk control at $\alpha = 0.1$, calibrated on half of the items (168 failures per split), holds false PASS at 9% on the held-out half with 0% false FAIL. The guarantee is for the pooled mix of failures, not for each type: under the same threshold the held-out false PASS is 21% on swaps (down from 49% at the default threshold), 1% on wrong items and 0% on random rearrangements. The ranking holds on near-misses alone (swap-only AUROC 0.96), so calibrating on swaps (72 per split) holds their false PASS at 9% at a false-FAIL cost of 8%. A high AUROC shows good ranking, not calibrated probabilities; what it buys here is that a recalibrated threshold yields much better decisions than the default one, with the miss rate controlled under exchangeability.

C. Memorized orders fill gaps and pull the reading

For Qwen3.5-4B the memorized-versus-arbitrary contrast matters little when the scene is informative and the order is listed: with the instruction given, swaps are reported in the instructed order 0% (rainbow) vs. 0% (random blocks) and 7% (alphabetical) vs. 12% (random tiles), and the verdict errors above are similar across all four families. The memorized order shows up elsewhere. Merely *mentioning* that the robot was asked for alphabetical order, without listing it, makes the model report 26% of swaps in alphabetical order and lowers its accuracy on random sequences from 100% to 66%. The errors move the answer toward the alphabet: positions agreeing with alphabetical order rise from 6% in the true sequences to 17% in the answers, and 3 of 24 random sequences come back as exactly A–H. The model partly sorts what it sees. Mentioning rainbow order has no such effect (0%), consistent with the alphabet being the stronger prior. And when the scene carries no information, the prior fills it in: on blank tiles the model reports A–H in 100% of cases even with a neutral prompt, and with a random order given it reports that order (100%). The verdict follows: asked PASS or FAIL about blank tiles, it answers PASS on 100% of the alphabetical and 100% of the random-order scenes (three each), even when made to list the tiles first. A success detector built on this model confirms an outcome it cannot see.

D. Across model size and family

Fig. 4 shows that the three effects vary differently with the model. *Near-miss blindness is shared by every model we test.* Each passes a large share of single swaps (95%, 65% and 49% for Qwen3.5-0.8B, 2B and 4B, 84% for Gemma 4 E2B), and most of the swaps each passes are ones it transcribes correctly with a neutral prompt in a separate call (88%, 82%, 99% and 78%). For Qwen3.5-0.8B and Gemma the reversed question shows no discrimination at this operating point ($J = -0.01$ and 0.01): their answers track the wording of the question more than the arrangement.

Stated instructions leak into perception in the weaker models. With the instruction in the prompt, Qwen3.5-0.8B and Gemma report most swaps as the instructed order, for memorized and random orders alike (Fig. 4a); within Qwen3.5 this falls steeply with size, to 0%–12% at 4B.

The memorized order bends perception even when nobody states it. With a neutral prompt that mentions neither a robot nor an order, Qwen3.5-0.8B, Qwen3.5-2B and Gemma report 40%, 38% and 64% of swapped alphabetical tiles in alphabetical order, against 0%, 0% and 0% for the same kind of swap in a random-order control. Only the 4B model does not show this (0%). Because this happens before any comparison, it also undermines the code-comparison fix on alphabetical tiles ($J = 0.80$, 0.74 and 0.66, against 1.00 at 4B). Rainbow order, which all models can recite (83%–100% text-only recall), never has this effect (0% for every model). Being able to recite an order is therefore not sufficient; we conjecture that what matters is how strongly the order is primed as a continuation, which is far stronger for the alphabet.

Uncertainty remains informative, but not equally. The verdict score ranks failures above successes for every model (pooled AUROC 0.87, 0.93, 0.98, 0.86), but its usefulness differs. Many of Gemma's failures receive a low score: 42% of failed arrangements score below 0.1, against 4% for Qwen3.5-4B (Fig. 3). Conformal risk control holds false PASS near the 10% target for every model (10%, 10%, 9%, 10%), but at a false-FAIL cost that ranges from 0% (Qwen3.5-4B) to 64% (Gemma), and for every model it leaves 21%–22% of swaps

TABLE I

DETECTION MINUS FALSE-ALARM RATE (J) OF FIVE WAYS TO VERIFY AN ARRANGEMENT. FOR EACH MODEL THE COLUMNS ARE BLOCKS IN RAINBOW ORDER (RB) AND WITH A RANDOM GOAL (RG), THEN TILES IN ALPHABETICAL ORDER (AL) AND WITH A RANDOM GOAL (RG). THE FIRST THREE ROWS ARE SINGLE MODEL CALLS; IN THE LAST TWO THE MODEL ONLY TRANSCRIBES AND CODE COMPARES WITH THE INSTRUCTION.

	Qwen3.5-4B				Gemma 4 E2B			
Strategy	Rb	Rg	Al	Rg	Rb	Rg	Al	Rg
Ask PASS/FAIL	0.67	0.74	0.73	0.69	0.51	0.78	0.33	0.60
Ask "any item out of place?"	0.94	0.99	0.70	0.73	0.00	0.00	0.00	0.02
List first, then verdict	1.00	0.97	0.96	0.86	0.73	0.77	0.87	0.76
Transcribe with order given, code compares	1.00	1.00	0.97	0.94	0.62	0.42	0.58	0.61
Transcribe neutrally, code compares	1.00	1.00	1.00	1.00	1.00	0.96	0.66	1.00

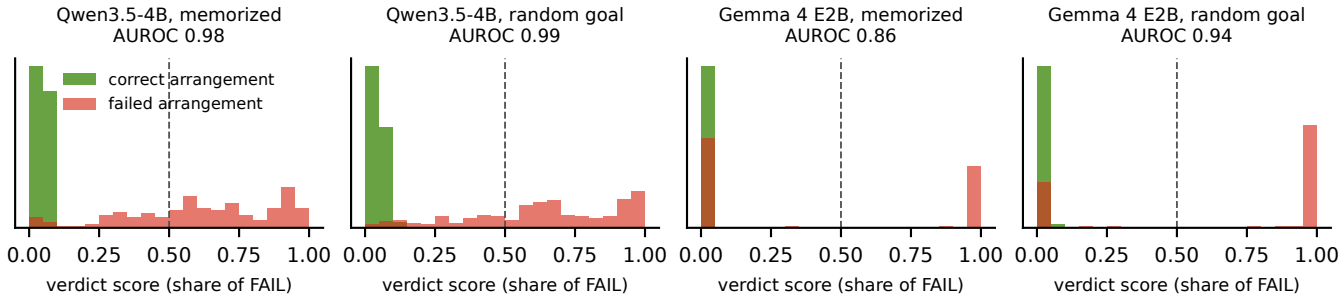


Fig. 3. Verdict score under the PASS/FAIL prompt for correct and failed arrangements, for Qwen3.5-4B and Gemma 4 E2B. "Memorized" pools the rainbow and alphabetical families, "random goal" the two random-order families. The dashed line is the threshold implied by greedy decoding; AUROC measures how well the score ranks failures above successes. Qwen3.5-4B's misses sit at intermediate scores, where a recalibrated threshold catches them; many of Gemma's sit near 0.

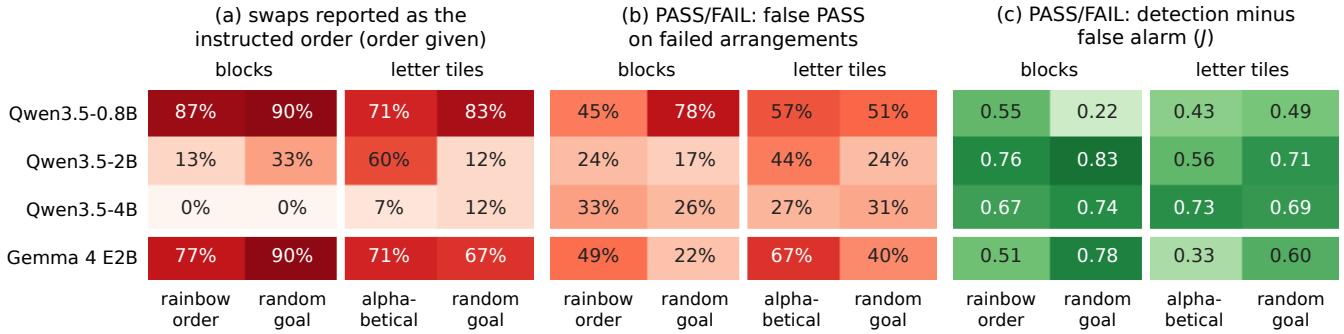


Fig. 4. All four models on the controlled benchmark. (a) Adjacent swaps transcribed as the instructed order when that order is given in the prompt. (b, c) The PASS/FAIL verdict: false PASS on failed arrangements, and detection minus false-alarm rate J. Columns pair each memorized order with its random-goal control.

passing. Calibrating on swaps alone is affordable only for Qwen3.5-4B; it costs the other models 71%–83% false FAIL.

E. Real robot episodes

Under this final-frame protocol the verdicts show little discrimination (Table II). Pooled over the five instruction types, J of the PASS/FAIL verdict lies between −0.07 and 0.06 for the four models and the AUROC of the verdict score between 0.45 and 0.51 (0.49–0.53 for the reversed question and 0.47–0.50 for describe-first). Each model instead applies a fixed bias: Qwen3.5-0.8B passes everything, Qwen3.5-4B passes 87% of the counterfactuals, and Qwen3.5-2B and Gemma fail most true instructions (84% and 95%). With a near-chance pooled AUROC, thresholding offers little: a conformal threshold at $\alpha = 0.1$ rejects 86%–93% of completed episodes. (A near-chance pooled AUROC does not rule out signal in every subgroup or at every threshold.)

Ground truth on a real frame can be disputed: the frames are small, the relevant object is sometimes occluded by the arm, a final frame may not show that the instructed action happened, and we rely on the dataset's segmentation. The pairing gives a check with weaker assumptions. Read as predicates on the final state, the true and the counterfactual instruction are mutually exclusive (a drawer cannot end both open and closed), so

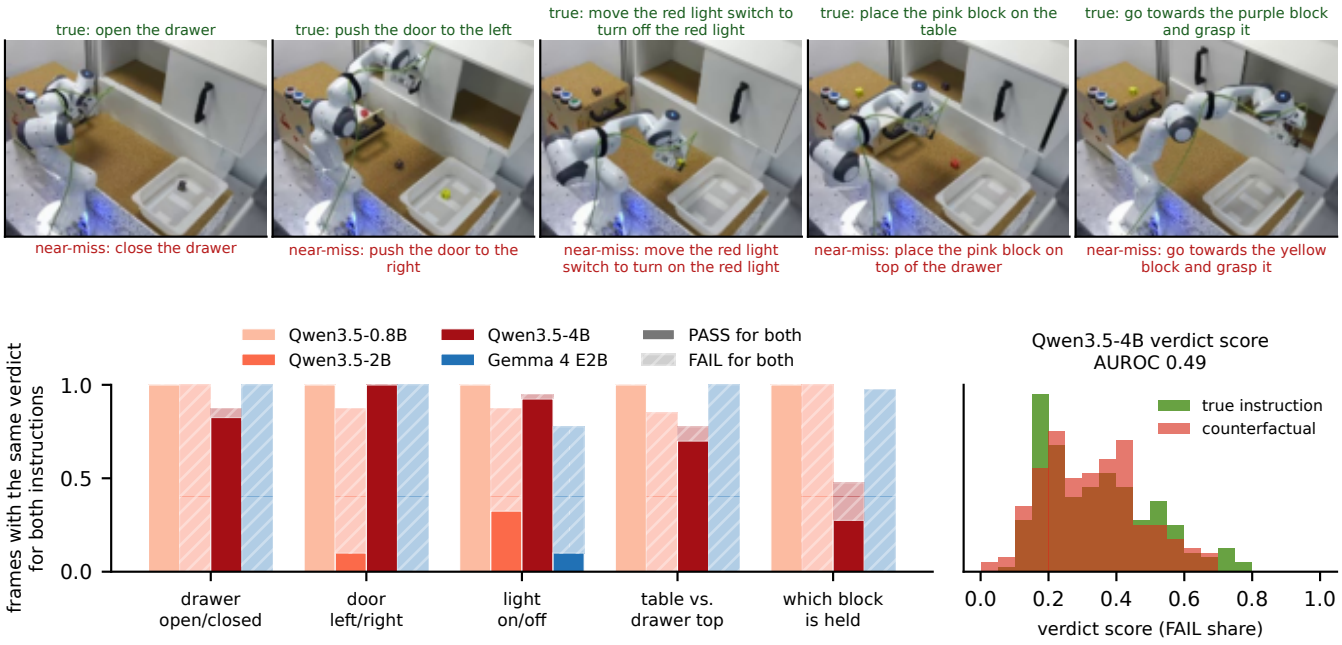


Fig. 5. Real-robot benchmark (TACO-Play). Top: one final frame per instruction type with its true instruction (green, correct verdict PASS) and the near-miss counterfactual (red, correct verdict FAIL). Bottom left: frames that receive the same PASS/FAIL verdict for both instructions, split into PASS for both (solid; two mutually exclusive outcomes certified) and FAIL for both (hatched; possibly insufficient evidence). Bottom right: Qwen3.5-4B's verdict score for true and counterfactual instructions.

TABLE II

REAL-ROBOT NEAR-MISS BENCHMARK, PASS/FAIL PROMPT. P/P AND F/F: FRAMES ON WHICH THE TRUE AND THE COUNTERFACTUAL INSTRUCTION BOTH RECEIVE PASS (TWO MUTUALLY EXCLUSIVE OUTCOMES CERTIFIED) OR BOTH RECEIVE FAIL. FP: FALSE PASS ON COUNTERFACTUAL INSTRUCTIONS. FF: FALSE FAIL ON TRUE ONES. J AND THE AUROC OF THE VERDICT SCORE, POOLED OVER THE FIVE INSTRUCTION TYPES. CONF. FF: FALSE FAIL OF A CONFORMAL THRESHOLD AT $\alpha = 0.1$.

Model	P/P	F/F	FP	FF	J	AUROC	conf. FF
Qwen3.5-0.8B	100%	0%	100%	0%	0.00	0.49	91%
Qwen3.5-2B	8%	84%	10%	84%	0.06	0.51	86%
Qwen3.5-4B	74%	7%	87%	20%	−0.07	0.49	93%
Gemma 4 E2B	2%	93%	4%	95%	0.01	0.45	91%

certifying both is a contradiction whatever the frame shows. The argument depends on this terminal-state reading: whether the robot *performed* an action is not always decidable from its final state. Rejecting both need not be a contradiction, since a frame may not show enough evidence for either. Qwen3.5-0.8B and Qwen3.5-4B certify both outcomes on 100% and 74% of the 200 frames, whereas Qwen3.5-2B and Gemma reject both on 84% and 93%, consistent with treating the frames as uninformative (Fig. 5). The describe-first prompt shows that the descriptions, too, depend on the instruction: for the same drawer or light frame, Qwen3.5-4B states a different state under the two instructions in 50% of the 50 frames where both descriptions state one (82% for Qwen3.5-0.8B, $n = 44$; 33% for Gemma, $n = 33$), for example "the drawer remains closed" when told to open it and "the drawer remains open" when told to close it. These results characterize this final-frame protocol; human checks of visibility and labels, and before/after frames, are needed before they can be read as a general statement about real-robot success detection.

F. Reasoning

On the reasoning subset (adjacent swaps under neutral light at the near zoom, 5–7 per family), enabling Qwen3.5-4B's thinking mode eliminated the observed false PASS on the 24 tested swaps: it falls from 20%, 40%, 71% and 43% to 0%, 0%, 0% and 0% (rainbow, random blocks, alphabetical, random tiles). The subset contains no correct arrangements, so specificity is untested (an always-FAIL detector would score the same), and thinking also changes the sampling settings and the computation, at a median of 434 generated tokens per answer instead of 11.

V. DISCUSSION

A decision failure more than a reading failure. For Qwen3.5-4B the error sits between reading and judging: in a separate call the model transcribes the near-miss correctly, and its verdict score ranks it as suspicious, but the one-word verdict reflects neither. This is the kind of failure uncertainty tools are designed for. The weaker models add a second, harder failure: the instruction, or a memorized order, changes what they read in the first place. Under our real-robot protocol the two cannot be separated; the verdict score shows weak pooled discrimination, so thresholding offers little, and two of the models certify mutually exclusive outcomes of the same frame.

Consequences for robot learning. A success detector that passes near-misses turns failed episodes into successes. It inflates evaluation, admits failed demonstrations into datasets filtered by VLM labels, and hands a reward-maximizing policy a target that a near-miss satisfies. Because the misses concentrate on near-misses, which typical evaluation sets rarely contain, average accuracy on typical episodes will not reveal them.

What helps. (i) *Calibrate the verdict.* With a modest labeled set, conformal risk control on the verdict score gives Qwen3.5-4B a miss rate controlled in expectation at almost no false-FAIL cost, because its score is graded. The guarantee covers the failure mix it is calibrated on, so near-misses should be calibrated on near-misses; and the share of failures with a low score (below 0.1: 4% for Qwen3.5-4B, 42% for Gemma) should be measured before any threshold is trusted. (ii) *Make the model look before it judges*, by listing the items first (or by reasoning, whose specificity we did not test). (iii) *Factorize*: transcribe with the instruction out of context and compare in code. This is error-free for Qwen3.5-4B, but it inherits whatever prior bends the transcription, as the alphabetical tiles show for the smaller models. (iv) *Evaluate on near-misses, with controls*: counterfactual instructions, and memorized versus arbitrary expectations over matched scene families, separate the failure modes that average accuracy conflates. Pairs of mutually exclusive instructions over the same frame are especially cheap: they need no labels, and a detector that certifies both outcomes of such a pair should not be trusted on a robot.

Limitations. The controlled scenes are synthetic; the real-robot benchmark covers one dataset, one camera and five instruction types, and relies on the dataset's annotation that each segment ends with its instruction completed. We study two visual domains and two memorized orders, one model family at three sizes and one model from a second family; the reasoning subset is small and has no correct-arrangement controls. The real-robot conclusions are bounded to a final-frame protocol on one dataset. Greedy decoding and one wording per framing leave paraphrase robustness untested. The tile prompts state that each tile shows a letter, which is false for the blank tiles by design; those scenes probe what the model fills in, not perception. Qwen3.5-2B gave no parseable verdict on 197 of its 432 list-first answers (coverage 54%); we score these as FAIL, which lowers its list-first J to 0.42. All other verdict prompts on the controlled benchmark have full coverage; on the real-robot benchmark only Qwen3.5-0.8B's describe-first answers lack a verdict (14 of 400, also scored as FAIL).

VI. Conclusion

Asked whether a robot did what it was told, small VLMs pass near-misses that they transcribe correctly when asked. For the strongest model we test, the evidence is in its own transcription and in its verdict score, and a recalibrated threshold or a look-before-you-judge prompt recovers most of it; for weaker models the instruction and memorized sequences leak into perception itself. Under a final-frame protocol on real robot data, all four models' verdict scores show weak pooled discrimination and two certify mutually exclusive outcomes for most frames, so none should be trusted there without validation. Uncertainty-aware world models for robotics should be evaluated on exactly these cases, where what the model expects and what the camera shows differ in one detail. Code, stimuli, every model output and the real-robot benchmark will be released.

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Appendix: prompts and settings

Prompts differ only in text; the image is attached before the text of every prompt.

Blocks

Neutral (no context). The image shows 6 colored blocks in a row. List the color of each block from left to right.
Allowed color names: blue, green, orange, pink, purple, red, white, yellow.
Answer with exactly 6 color names separated by commas, and nothing else.

Scene named. The image shows a table after a robot arm has arranged 6 colored blocks in a row. List the color of each block from left to right.
Allowed color names: blue, green, orange, pink, purple, red, white, yellow.
Answer with exactly 6 color names separated by commas, and nothing else.

Expectation mentioned. The image shows a table after a robot arm was instructed to arrange 6 colored blocks in rainbow order. List the color of each block from left to right exactly as they appear in the image.
Allowed color names: blue, green, orange, pink, purple, red, white, yellow.
Answer with exactly 6 color names separated by commas, and nothing else.

Expectation given. The image shows a table after a robot arm was instructed to arrange 6 colored blocks in a row. The instruction was to place the blocks in rainbow order, from left to right: red, orange, yellow, green, blue, purple. List the color of each block from left to right exactly as they appear in the image.
Allowed color names: blue, green, orange, pink, purple, red, white, yellow.
Answer with exactly 6 color names separated by commas, and nothing else.

PASS/FAIL verification. The image shows a table after a robot arm was instructed to arrange 6 colored blocks in a row. The instruction was to place the blocks in rainbow order, from left to right: red, orange, yellow, green, blue, purple. Inspect the table: are the blocks in exactly this order? Answer with one word: PASS if the order is correct, or FAIL if any block is out of place.

Deviation question (YES = deviation). The image shows a table after a robot arm was instructed to arrange 6 colored blocks in a row. The instruction was to place the blocks in rainbow order, from left to right: red, orange, yellow, green, blue, purple. Inspect the table: is any block out of place compared with this order? Answer with one word: YES if any block is out of place, or NO if every block is in the correct position.

List first, then PASS/FAIL. The image shows a table after a robot arm was instructed to arrange 6 colored blocks in a row. The instruction was to place the blocks in rainbow order, from left to right: red, orange, yellow, green, blue, purple. First, on one line, list the color of each block from left to right exactly as it appears in the image, as 6 color names separated by commas. Allowed color names: blue, green, orange, pink, purple, red, white, yellow.
Then, on a second line, compare your list with the instruction and answer with one word: PASS if the order is correct, or FAIL if any block is out of place.

Random-goal variant: identical except that "in rainbow order." becomes "in a specific order."; "The instruction was to place the blocks in rainbow order, from left to right: red, orange, yellow, green, blue, purple." becomes "The instruction was to place the blocks, from left to right, in this order: ⟨*instructed order*⟩." (the order is sampled per image).

Letter tiles

Neutral (no context). The image shows 8 letter tiles in a row. Read the letter on each tile from left to right.
Each tile shows one capital letter.
Answer with exactly 8 capital letters separated by commas, and nothing else.

Scene named. The image shows a table after a robot arm has arranged 8 letter tiles in a row. Read the letter on each tile from left to right.
Each tile shows one capital letter.
Answer with exactly 8 capital letters separated by commas, and nothing else.

Expectation mentioned. The image shows a table after a robot arm was instructed to arrange 8 letter tiles in alphabetical order. Read the letter on each tile from left to right exactly as they appear in the image.
Each tile shows one capital letter.
Answer with exactly 8 capital letters separated by commas, and nothing else.

Expectation given. The image shows a table after a robot arm was instructed to arrange 8 letter tiles in a row. The instruction was to place the tiles in alphabetical order, from left to right: A, B, C, D, E, F, G, H. Read the letter on each tile from left to right exactly as they appear in the image.
Each tile shows one capital letter.
Answer with exactly 8 capital letters separated by commas, and nothing else.

PASS/FAIL verification. The image shows a table after a robot arm was instructed to arrange 8 letter tiles in a row. The instruction was to place the tiles in alphabetical order, from left to right: A, B, C, D, E, F, G, H. Inspect

the table: are the tiles in exactly this order? Answer with one word: PASS if the order is correct, or FAIL if any tile is out of place.

Deviation question (YES = deviation). The image shows a table after a robot arm was instructed to arrange 8 letter tiles in a row. The instruction was to place the tiles in alphabetical order, from left to right: A, B, C, D, E, F, G, H. Inspect the table: is any tile out of place compared with this order? Answer with one word: YES if any tile is out of place, or NO if every tile is in the correct position.

List first, then PASS/FAIL. The image shows a table after a robot arm was instructed to arrange 8 letter tiles in a row. The instruction was to place the tiles in alphabetical order, from left to right: A, B, C, D, E, F, G, H. First, on one line, read the letter on each tile from left to right exactly as it appears in the image, as 8 capital letters separated by commas.
Then, on a second line, compare your list with the instruction and answer with one word: PASS if the order is correct, or FAIL if any tile is out of place.

Random-goal variant: identical except that "in alphabetical order." becomes "in a specific order."; "The instruction was to place the tiles in alphabetical order, from left to right: A, B, C, D, E, F, G, H." becomes "The instruction was to place the tiles, from left to right, in this order: ⟨*instructed order*⟩." (the order is sampled per image).

Real-robot prompts

PASS/FAIL. The image shows a robot workspace after the robot was instructed to: "⟨*instruction*⟩". Did the robot complete the instruction? Answer with one word: PASS if it did, or FAIL if it did not.

Reversed question (YES = failure). The image shows a robot workspace after the robot was instructed to: "⟨*instruction*⟩". Did the robot fail to complete the instruction? Answer with one word: YES if it failed, or NO if it completed it.

Describe first, then PASS/FAIL. The image shows a robot workspace after the robot was instructed to: "⟨*instruction*⟩". First, in one sentence, describe the state of the objects the instruction refers to, as they appear in the image. Then, on a second line, answer with one word: PASS if the instruction was completed, or FAIL if it was not.

The instruction is the segment's own annotation (correct verdict PASS) or its counterfactual (correct verdict FAIL).

Models and generation settings

Hugging Face weights (the same revisions for both benchmarks): `Qwen/Qwen3.5-0.8B` at `2fc06364715b`, `Qwen/Qwen3.5-2B` at `15852e8c1636`, `Qwen/Qwen3.5-4B` at `851bf6e806ef`, `google/gemma-4-E2B-it` at `3e22461f65e8`. All models run in bf16 with PyTorch 2.14.0 and Transformers 5.17.0, with each model's default image processing. Decoding is greedy with reasoning disabled, with at most 96 new tokens for transcriptions, 16 for one-word verdicts, 160 for list first, 120 for describe first and 96 for the text-only recall probes. The reasoning subset samples with temperature 0.6, top-p 0.95 and top-k 20, with at most 8192 new tokens. The verdict is the last occurrence of the requested verdict words (case-insensitive), falling back to the other pair (PASS/FAIL or YES/NO); an answer with neither has no verdict. The supplementary material contains the code, the stimuli, and every prompt and raw answer, with verdict scores and token log-probabilities where recorded.